

Day 13 Of Communication System Class

8.3 Error correction codes:

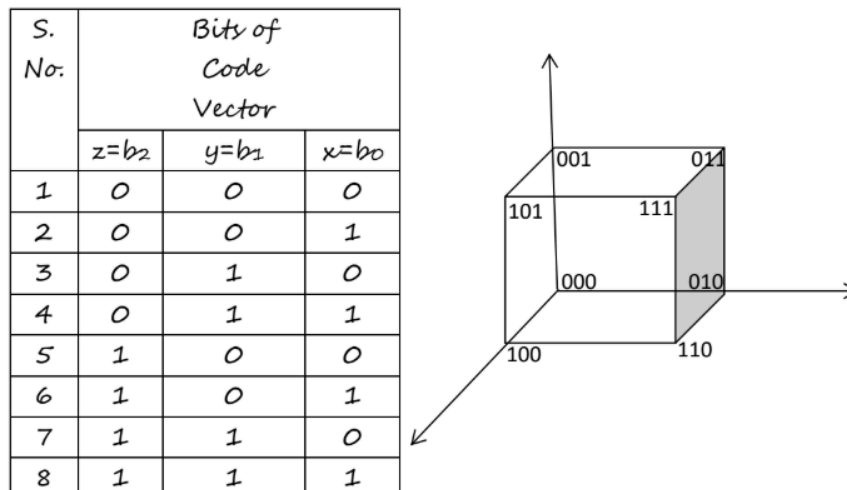
8.3.1 Linear block codes

8.3.2 Hamming codes

8.3.3 Convolutional code

53. What is code vector? Visualize a 3-bit codeword as code vector.

-Visualization of 'n' bit codeword in an n-dimensional space as a vector whose elements or coordinates are the bits in the codeword.



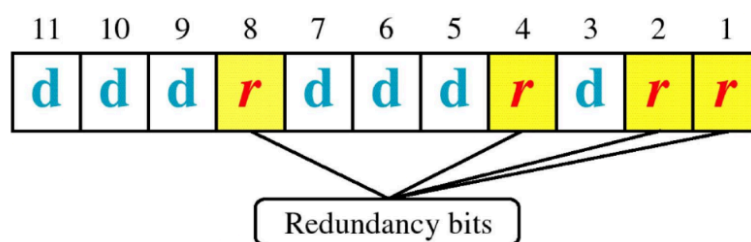
54. What are linear block codes?

-For the block of 'k' message bits, (n-k) redundant check bits are added. This gives (n,k) block codes with 'n' bits in the o/p of the channel encoder

-Block codes are known as linear if the mod-2 addition of any two code vectors produces another code vector.

8.3.2 Hamming codes

- It is constructed by adding a no. of parity bits to each group of n-bit information, or message in such a way so as to be able to locate the bit position in which error occurs.
- Values (0 or 1) are assigned to the parity bits so as to make the hamming code have either even parity or odd parity & when an error occurs, the position no. will take on the value assigned to the location of the erroneous bit.



- Hamming code is an error-correcting code used to ensure data accuracy during transmission or storage.
- Hamming code detects and corrects the errors that can occur when the data is moved or stored from the sender to the receiver.
- This simple and effective method helps improve the reliability of communication systems and digital storage.
- It adds extra bits to the original data, allowing the system to detect and correct single-bit errors.
- It is a technique developed by [Richard Hamming](#) in the 1950s.

Algorithm of Hamming Code

Hamming Code is simply the use of extra parity bits to allow the identification of an error.

- **Step 1:** Write the bit positions starting from 1 in binary form (1, 10, 11, 100, etc).
- **Step 2:** All the bit positions that are a power of 2 are marked as parity bits (1, 2, 4, 8, etc).
- **Step 3:** All the other bit positions are marked as data bits.
- **Step 4:** Each data bit is included in a unique set of parity bits, as determined its bit position in binary form:

Q. With suitable assumptions for a code word of $C(7,4)$, demonstrate how Hamming Code detects an error. [2081 Ashwin]

Q. Generate a hamming code for 10101100 using even parities, and demonstrate how it corrects a single bit error. [2082 Chaitra]